

Assessment-2

Data Interpretation

Q1)

1. R program:

```
library(ggplot2)

data <- data.frame(
  Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
  Sales = c(120, 90, 150, 80, 135, 100),
  Profit = c(18, 12, 25, 8, 22, 16),
  Category =
c("Electronics", "Clothing", "Electronics",
  "Furniture", "Furniture", "Clothing")
)

ggplot(data, aes(x = Sales, y = Profit,
  color = Category,
  size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(title = "Sales vs Profit",
    x = "Sales (Lakhs)",
    y = "Profit (%)") +
  theme_minimal()
```

2. Aesthetic Mapping

- **X-axis:** Sales
- **Y-axis:** Profit
- **Color:** Category
- **Size:** Sales

3. Interpretation

- Profit generally increases as Sales increase.
- Electronics products have the highest sales and profits.
- Furniture has moderate performance.
- Clothing has comparatively lower sales and profits.

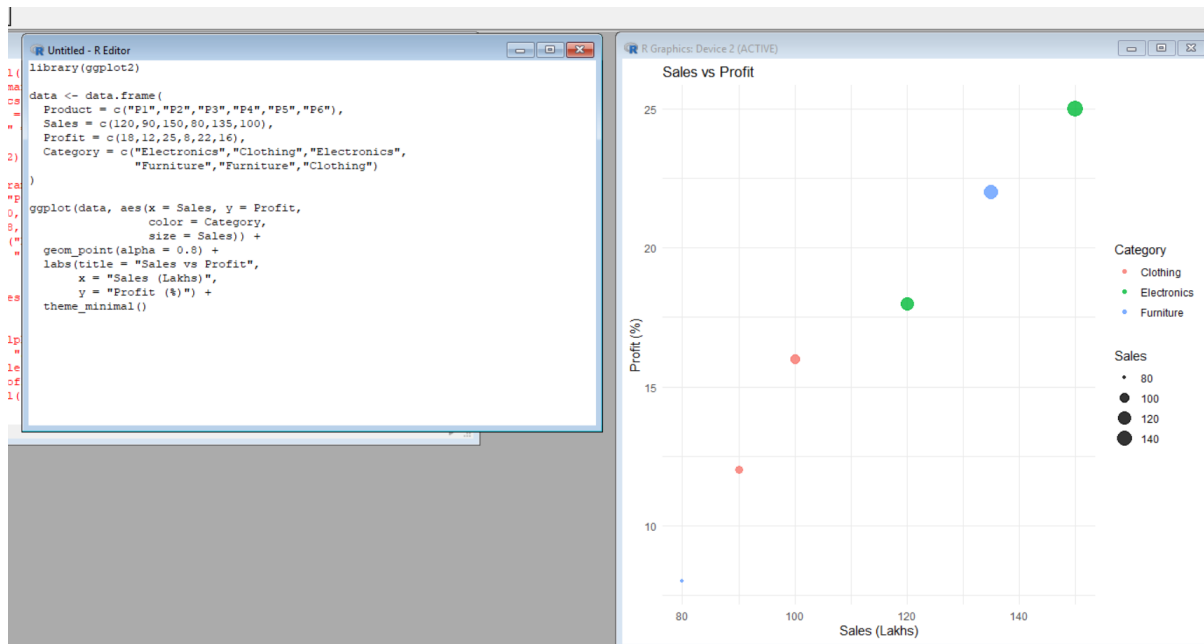
4. Scale Types

Variable		Scale
➤ Sales	➔	Continuous
➤ Profit	➔	Continuous
➤ Category	➔	Categorical (Discrete)

5. Alternative Visualization

Bubble Chart or Grouped Bar Chart.

Output:



Q2)

1. 1st R program for Cartesian Coordinate System:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
Direction=c("N","NE","E","SE","S","SW","W",
", "NW"),
Frequency=c(18,12,20,15,22,11,16,14)
)
```

```
ggplot(data, aes(Direction, Frequency))+  
geom_bar(stat="identity", fill="steelblue")+  
theme_minimal()+  
labs(title="Direction Frequency")
```

2nd R program for Polar Coordinate System:
library(ggplot2)

```
ggplot(data, aes(Direction, Frequency,  
fill=Direction))+  
geom_bar(stat="identity")+  
coord_polar()+  
theme_minimal()+  
labs(title="Direction Frequency (Polar)")
```

3. Comparison

- Cartesian chart is easier for comparing frequencies.
- Polar chart represents directions naturally.

4. Best Coordinate System

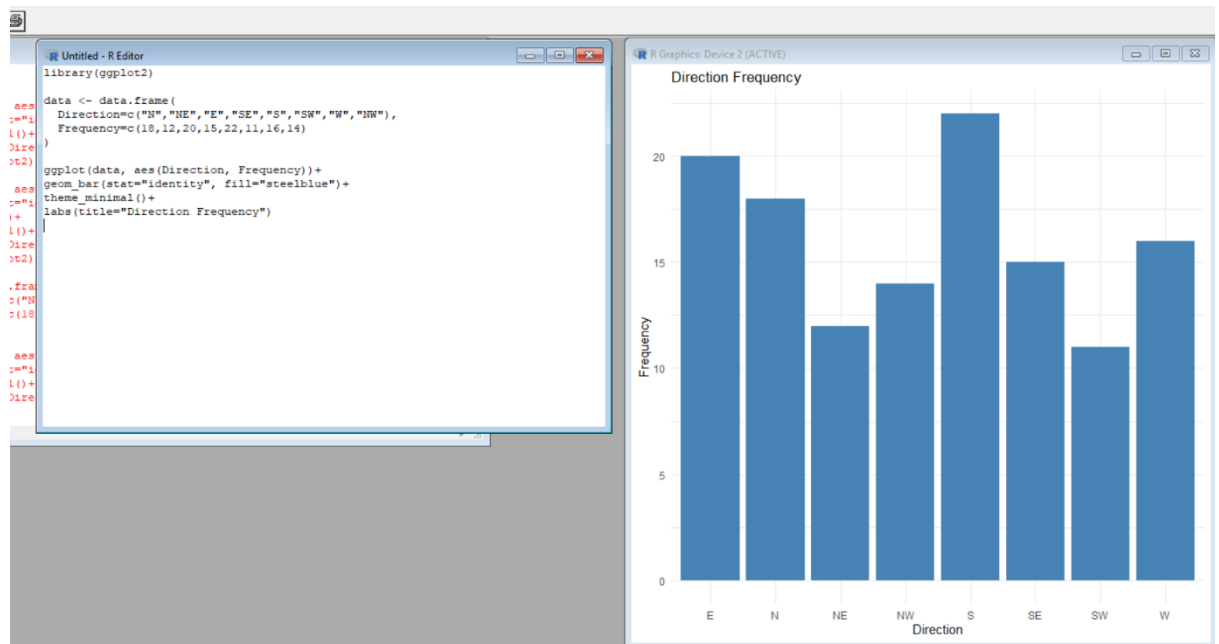
- Polar Coordinate System

5. Justification

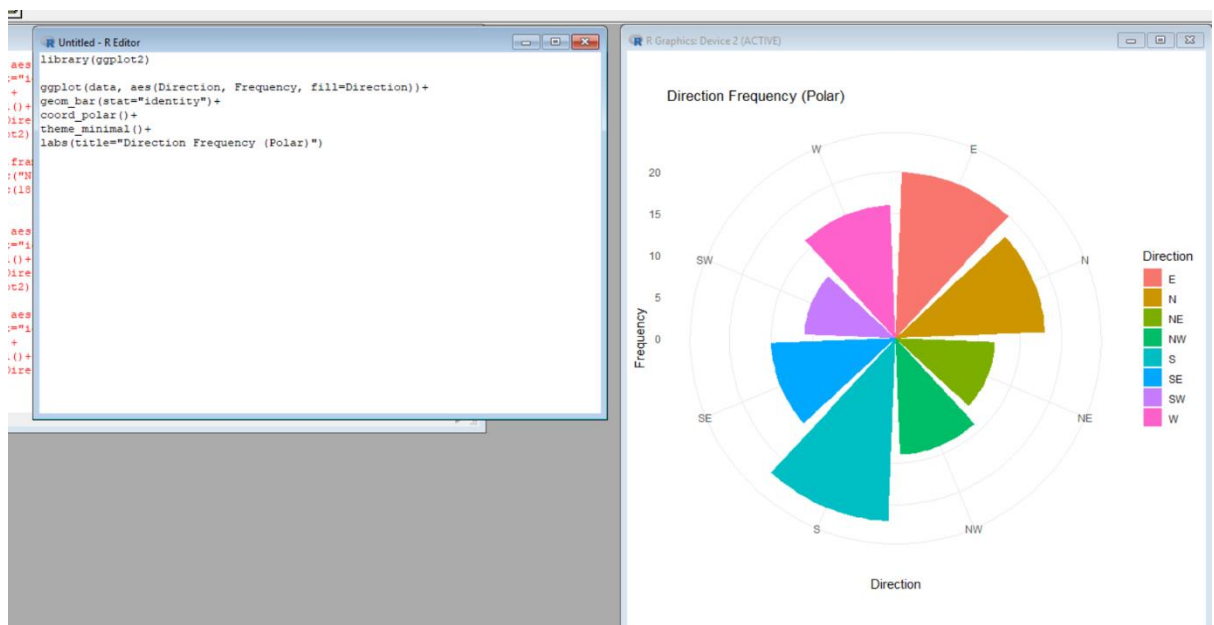
- Directional data is circular, making the polar chart more intuitive and easier to interpret.

Outputs:

1. Cartesian Coordinate System:



2. Polar Coordinate System:



Q3)

1. R program:

```
library(ggplot2)
```

```
data <- data.frame(  
  Student=c("A","B","C","D","E","F"),  
  Marks=c(45,58,72,81,67,90)  
)
```

```
ggplot(data,aes(Student,Marks,fill=Marks))+  
  geom_bar(stat="identity")+  
  scale_fill_viridis_c()+  
  theme_minimal()+  
  labs(title="Student Marks")
```

2. Highest and Lowest Marks

- Highest: Student F (90)
- Lowest: Student A (45)

3. Why Sequential Color Scale?

- Sequential colors represent increasing numerical values from low to high, making comparison easier.

4. Color-blind Friendly Palette

- Viridis

5. Difference

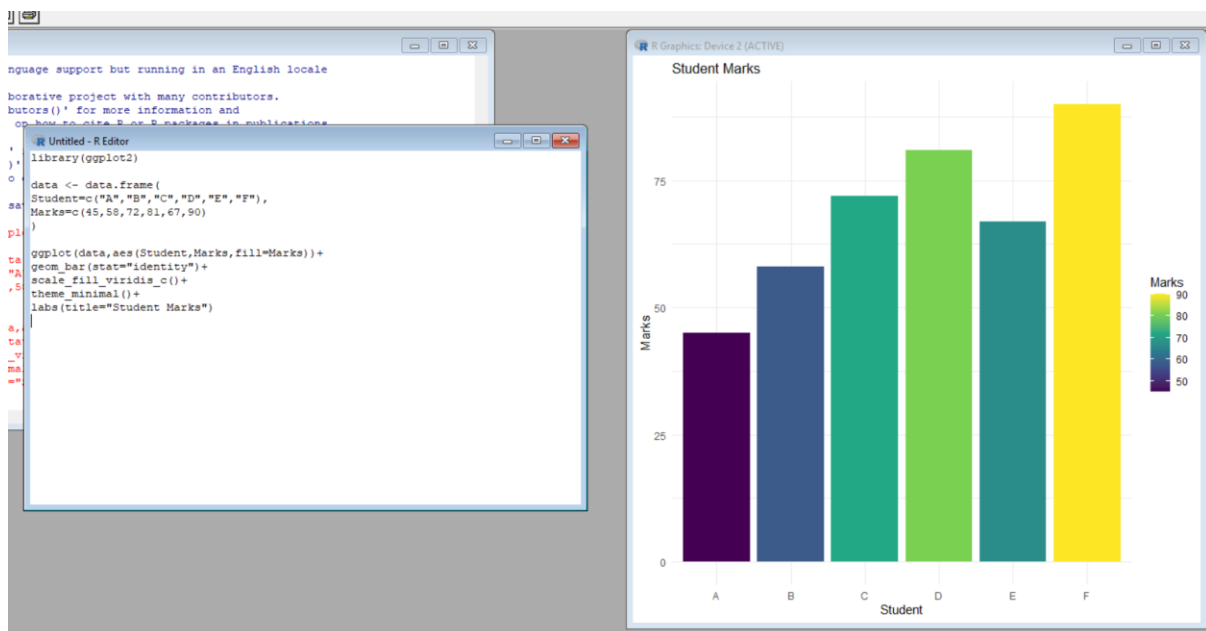
Sequential

- Used for numeric data
- Gradient colors
- Shows order

Categorical

- Used for categories
- Distinct colors
- No order

Output:



Q4)

1. R program

```
library(ggplot2)
```

```
data <- data.frame(
```

```
Region=c("North","South","East","West","  
Central"),  
Revenue=c(820,760,790,1010,770)  
)
```

```
data$Color <-  
ifelse(data$Revenue==max(data$Revenue),"r  
ed","gray")
```

```
ggplot(data,aes(Region,Revenue,fill=Color))+  
geom_bar(stat="identity")+  
scale_fill_identity()+  
theme_minimal()+  
labs(title="Regional Revenue")
```

2. Importance of Highlighting

- Draws attention to key values.
- Makes comparisons easier.
- Helps identify trends quickly.

3. Highest Performing Region

- West (₹1010 Crores)

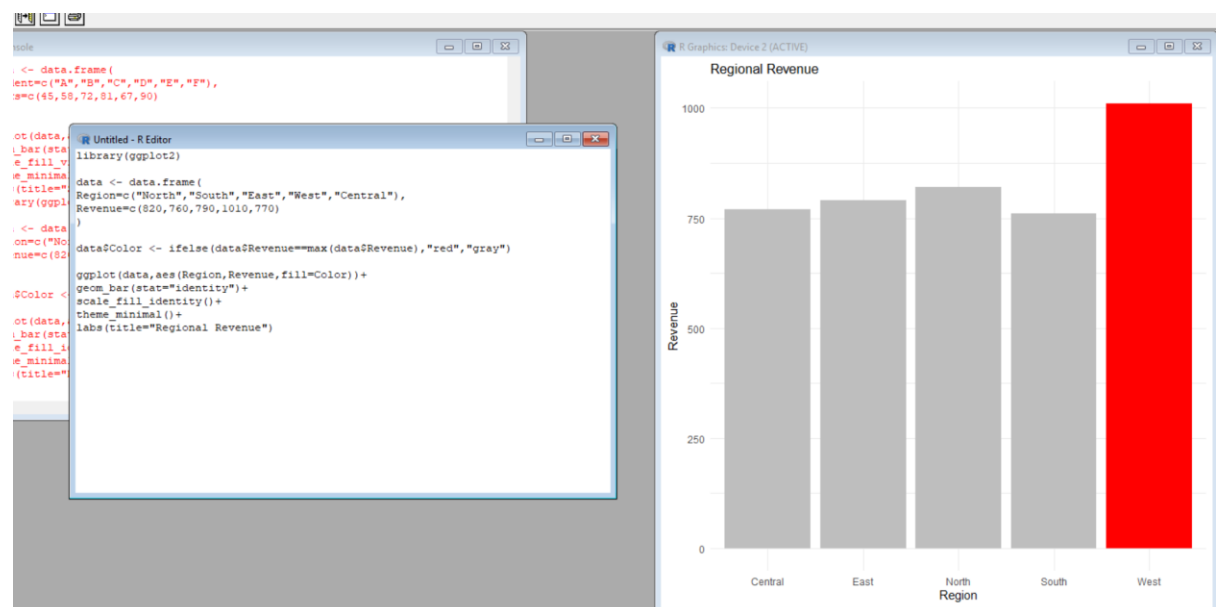
4. Effect of Excessive Colors

- Makes charts confusing.
- Reduces readability.
- Distracts viewers.

5. Alternative Highlighting Technique

- Add labels
- Use larger bars
- Add annotations

Outputs:



Q5)

1. R program:

library(ggplot2)

data <- data.frame(

City=c("A","B","C","D","E"),

```
Population=c(8.2,5.6,9.4,6.8,4.9),  
AQI=c(110,180,90,240,150),  
Rainfall=c(850,430,1200,520,680),  
Region=c("North","East","West","South",  
"East")  
)
```

```
ggplot(data,  
aes(x=Population,  
y=AQI,  
size=Rainfall,  
color=Region))+  
geom_point(alpha=0.8)+  
theme_minimal()+  
labs(title="Population vs AQI")
```

2. Aesthetic Mapping

- X-axis: Population
- Y-axis: AQI
- Bubble Size: Rainfall
- Color: Region

3. Interpretation

- Cities with higher populations generally tend to have higher AQI.
- Rainfall varies across cities and is represented by bubble size.

- Regions are distinguished using different colors.

4. Variable Types

- Continuous Variables
- Population
- AQI
- Rainfall
- Categorical Variable
- Region

5. Improvement

- Add city labels using `geom_text()`.
- Increase font size and use a clear legend for better readability.

Output:

